



Fundamentals of Optics

Lecturer

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Room 239, South Tower, College of Engineering

Rules

- Final exam: 30%
- Mid-term exam: 30%
- Homework: 25%
- Attendance: 15%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

CHAPTER 8

INTERFERENCE

Interference in daily life



Optical interference corresponds to the interaction of two or more lightwaves yielding a resultant irradiance that deviates from the sum of the component irradiances.

Wavefront splitting (分波阵面): portions of the primary wavefront are used either directly as sources to emit secondary waves or in conjunction with optical devices to produce virtual source of secondary waves. These secondary waves are then brought together to interference.

Example: Young's double slit experiment

Amplitude splitting (分振幅) : the primary wave itself is divided into two segments, which travel different paths before recombining and interfering.

Example: fringes of equal inclination

8.1 General Considerations

Principle of superposition:

The electric field intensity at a point in space:

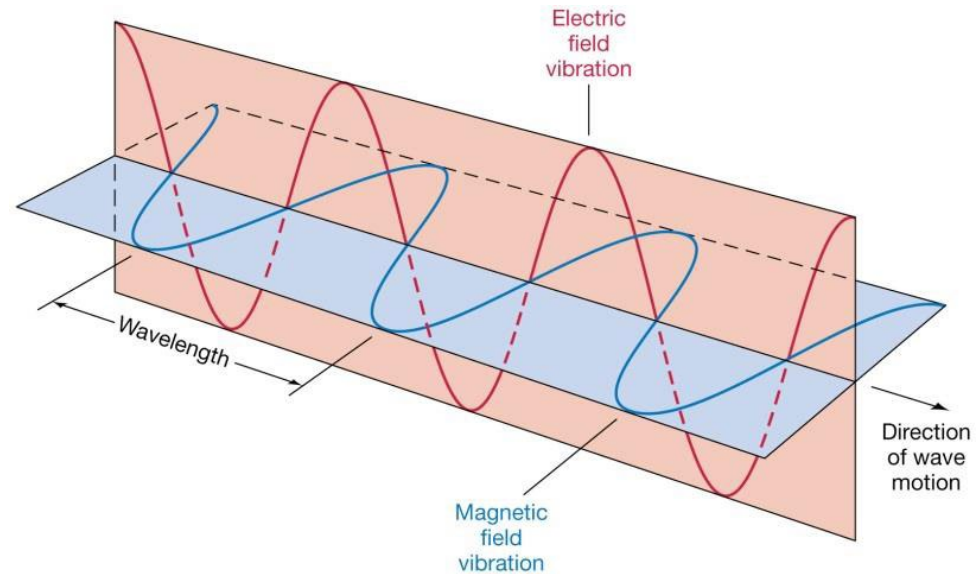
$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$$

The frequency of optical disturbance:

$$4.3 \times 10^{14} \text{ Hz to } 7.5 \times 10^{14} \text{ Hz}$$

The actual field is very difficult to detect, but the irradiance can be measured directly with sensors.

The irradiance:
$$I = \epsilon v \left\langle \vec{E}^2 \right\rangle_T$$



Electromagnetic field

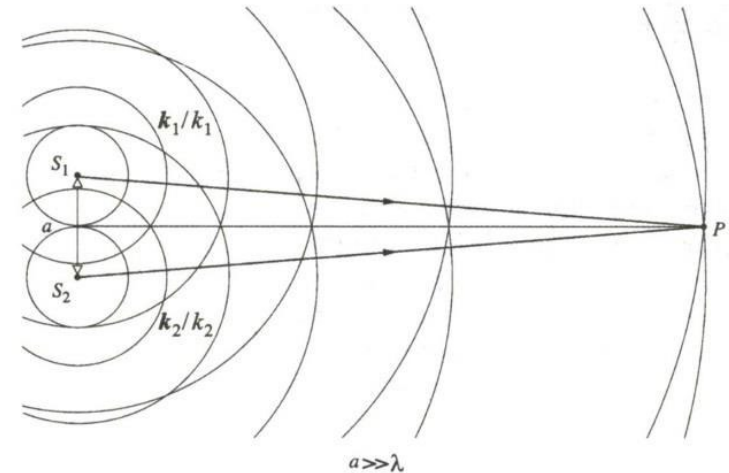
Case: consider two point sources emitting monochromatic waves of the same frequency in the same homogeneous medium;

- 1) $a(\text{the separation}) \gg \lambda(\text{the wavelength})$;
- 2) point of observation P is far from the source.

Two linearly polarized waves:

$$\vec{E}_1 = \vec{E}_{01} \cos(\vec{k}_1 \cdot \vec{r} - \omega t + \epsilon_1)$$

$$\vec{E}_2 = \vec{E}_{02} \cos(\vec{k}_2 \cdot \vec{r} - \omega t + \epsilon_2)$$



Irradiance at P (neglect the constant terms):

$$I = \langle \vec{E}^2 \rangle_T = \langle \vec{E} \cdot \vec{E} \rangle_T = \langle (\vec{E}_1 + \vec{E}_2) \cdot (\vec{E}_1 + \vec{E}_2) \rangle_T$$

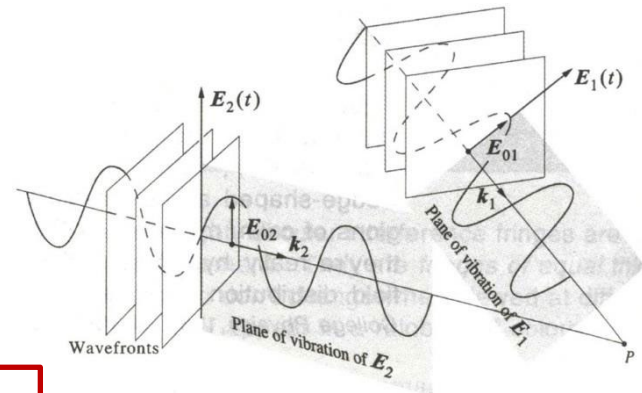
$$= \langle (\vec{E}_1^2 + \vec{E}_2^2 + 2\vec{E}_1 \cdot \vec{E}_2) \rangle_T = I_1 + I_2 + I_{12}$$

$$I_1 = \langle \vec{E}_1^2 \rangle_T$$

$$I_2 = \langle \vec{E}_2^2 \rangle_T$$

$$I_{12} = 2 \langle \vec{E}_1 \cdot \vec{E}_2 \rangle_T$$

Interference term



Interference term

$$I_{12} = 2 \left\langle \vec{E}_1 \cdot \vec{E}_2 \right\rangle_T = \vec{E}_{01} \cdot \vec{E}_{02} \cos \left(\vec{k}_1 \cdot \vec{r} - \vec{k}_2 \cdot \vec{r} + \epsilon_1 - \epsilon_2 \right) = \vec{E}_{01} \cdot \vec{E}_{02} \cos \delta$$

δ : phase difference arising from a combined path length and initial phase angle difference.

Case I: when $\vec{E}_{01}$ and $\vec{E}_{02}$ are perpendicular, $I_{12} = 0$, $I = I_1 + I_2$

Two such orthogonal polarized states will combine to yield an right-handed circularly polarized, left-handed circularly polarized, linearly polarized or elliptical polarized light (chapter 7), but the flux-density distribution will be unaltered.

Case II: when $\vec{E}_{01}$ and $\vec{E}_{02}$ are parallel

$$I_1 = \langle \vec{E}_1^2 \rangle_T = \frac{E_{01}^2}{2} \quad I_2 = \langle \vec{E}_2^2 \rangle_T = \frac{E_{02}^2}{2}$$

$$I_{12} = \vec{E}_{01} \cdot \vec{E}_{02} \cos \delta = 2\sqrt{I_1 I_2} \cos \delta$$

$$\text{Total irradiance: } I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

Total constructive interference: $\delta=0, \pm 2\pi, \pm 4\pi \dots$

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

Constructive interference: $0 < \cos \delta < 1$

$$I_1 + I_2 < I < I_{max}$$

When $\cos \delta = 0$, $I = I_1 + I_2$

Destructive interference: $0 > \cos \delta > -1$

$$I_1 + I_2 > I > I_{min}$$

Total destructive interference: $\delta=0, \pm 3\pi, \pm 5\pi \dots$

$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

Special case: $E_{01} = E_{02}$,
let $I_1 = I_2 = I_0$

$$I = 2I_0(1 + \cos \delta)$$

$$= 4I_0 \cos^2 \delta / 2$$

$$I_{max} = 4I_0 \text{ and } I_{min} = 0$$

Spherical waves emitted by S_1 and S_2

$$\vec{E}_1(r_1, t) = \vec{E}_{01}(r_1) \exp[i(kr_1 - \omega t + \epsilon_1)]$$

$$\vec{E}_2(r_2, t) = \vec{E}_{02}(r_2) \exp[i(kr_2 - \omega t + \epsilon_2)]$$

r_1, r_2 : the radii of the spherical wavefronts overlapping at P

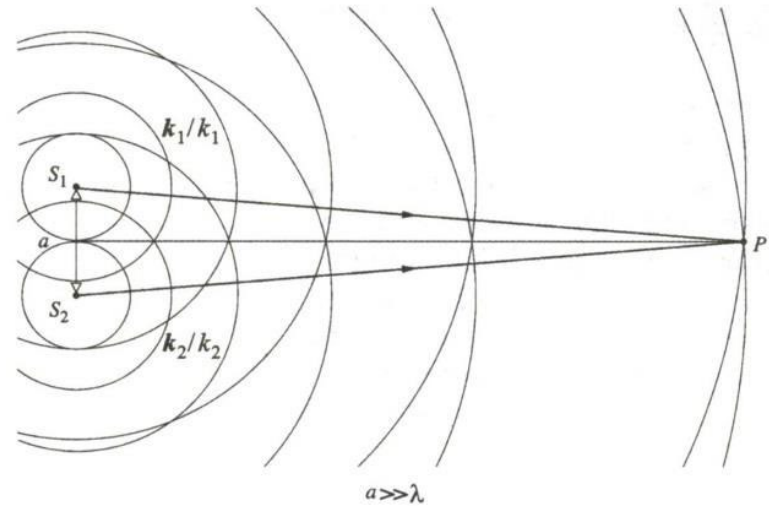
Irradiance for spherical waves:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

$$\delta = k(r_1 - r_2) + (\epsilon_1 - \epsilon_2)$$

When the separation (a) is small in comparison with r_1 and r_2 , and the interference region is also small, $\vec{E}_{01}, \vec{E}_{02}$ is considered independent of position, and assume that $\vec{E}_{01} = \vec{E}_{02}$

$$I = 4I_0 \cos^2 \frac{1}{2} [k(r_1 - r_2) + (\epsilon_1 - \epsilon_2)]$$



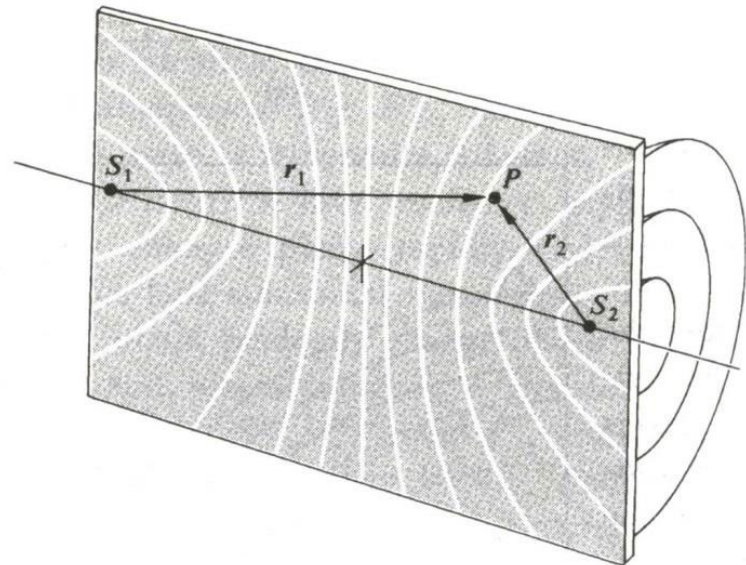
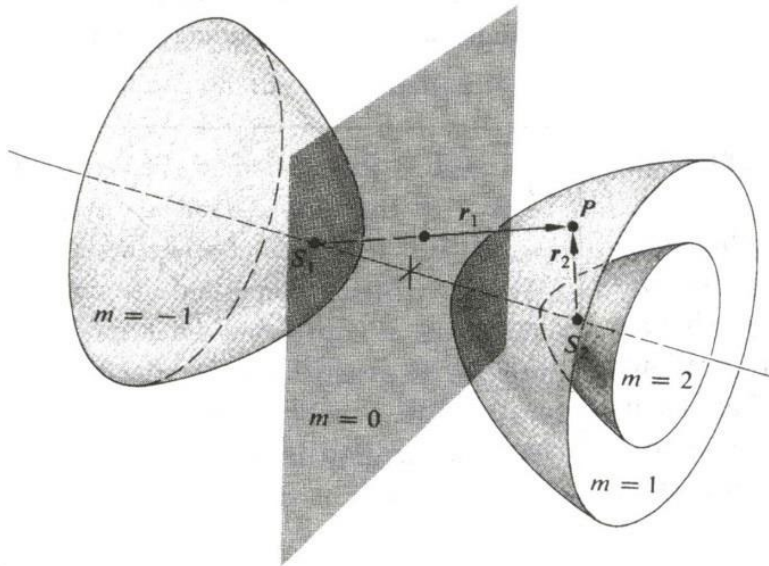
$$I = 4I_0 \cos^2 \frac{1}{2} [k(r_1 - r_2) + (\epsilon_1 - \epsilon_2)]$$

Irradiance maxima: $\delta = 2\pi m; m = 0, \pm 1, \pm 2, \dots$

$$(r_1 - r_2) = \frac{[2\pi m + (\epsilon_1 - \epsilon_2)] \text{ In-phase } 2\pi m}{k} = m\lambda$$

Irradiance minima: $\delta = (2m+1)\pi; m = 0, \pm 1, \pm 2, \dots$

$$(r_1 - r_2) = \frac{[(2m+1)\pi + (\epsilon_1 - \epsilon_2)] \text{ Out-phase } (2m+1)\pi}{k} = \left(m + \frac{1}{2}\right)\lambda$$



8.2 Conditions for Interference

Condition I: The same frequency

The interference term for two waves with different frequency:

$$\begin{aligned} & 2 \cos(\omega_1 t + \varphi_1) \cos(\omega_2 t + \varphi_2) \\ &= \cos[(\omega_1 + \omega_2)t + \varphi_1 + \varphi_2] + \cos[(\omega_1 - \omega_2)t + \varphi_1 - \varphi_2] \end{aligned}$$

The I_{12} averages to zero during the detection interval.

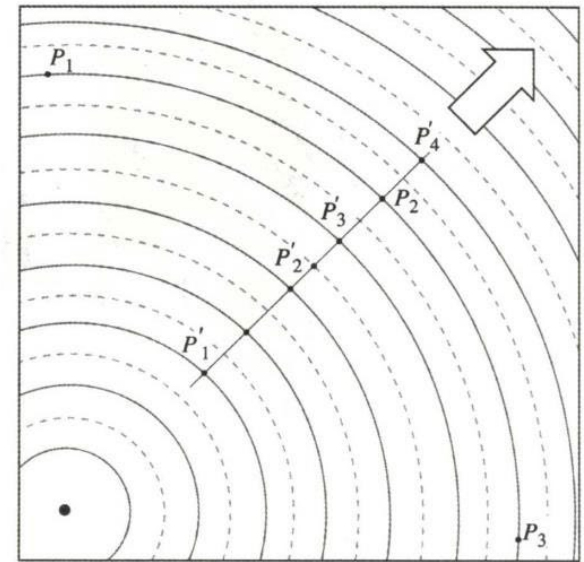
For white light:

The reds will interference with red, and the blues with blues.

Condition II: coherence

Conventional quasimonochromatic sources produce light that is a mix of photon wavetrains.

Coherent time and length: average time interval and corresponding spatial extent during which the lightwave oscillates in a predictable way.

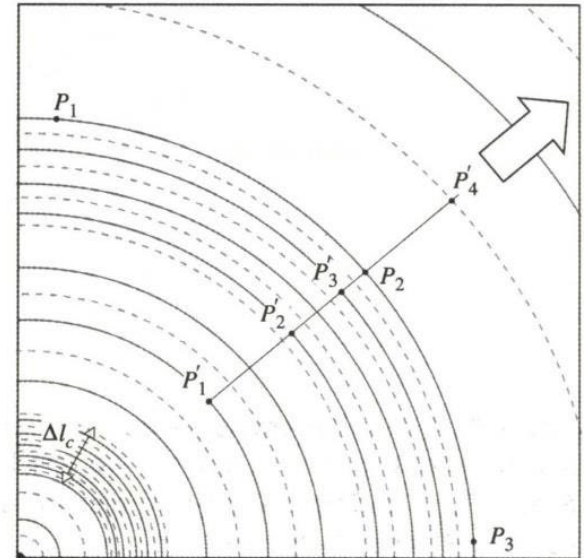


(a)

The waves display both forms of coherence perfectly

Temporal coherence: the interval over which the lightwave resembles a sinusoid before it randomly changes phase (about 10 ns).

Spatial coherence: P_1, P_2, P_3 lie on the same wavefront in both cases, and the behaviors of waves at these points is completely correlated.



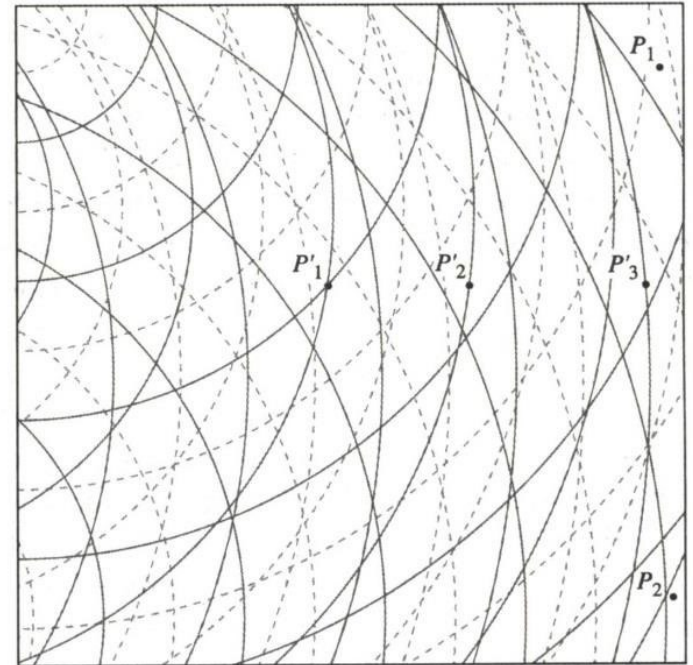
(b)

There is complete spatial coherence but only partial temporal coherence

Broad source (compose of many widely spaced point sources)

Each point source changes phase rapidly and randomly emitting 10-ns long sinusoidal wavetrains.

- ❑ The disturbance at P_1' , P_2' , and P_3' is only correlated for a time less than 10ns;
- ❑ The lateral point such as P_1 and P_2 is completely uncorrelated depending on the size of the source.

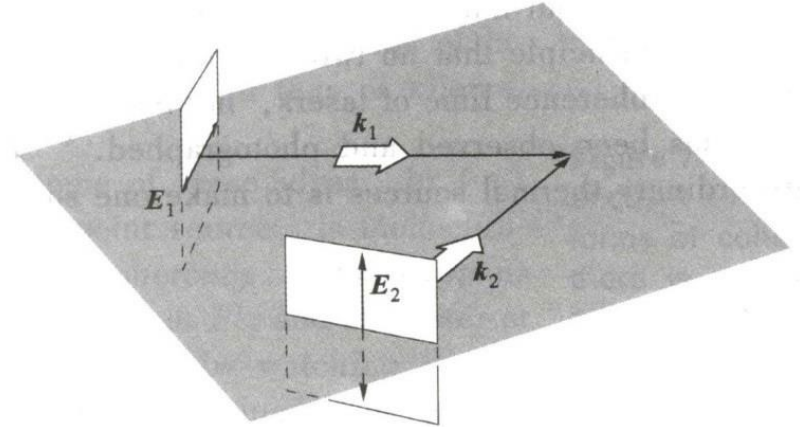
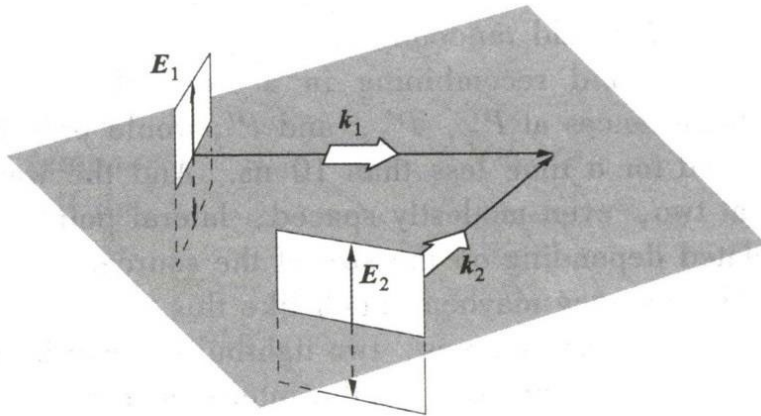


So it is impractical to observe interference for two sources, which maintain a constant relative phase for a time less than Δt .

Solution : one source serve to produce two coherent secondary sources.

Interference term

$$I_{12} = \vec{E}_{01} \cdot \vec{E}_{02} \cos \delta$$

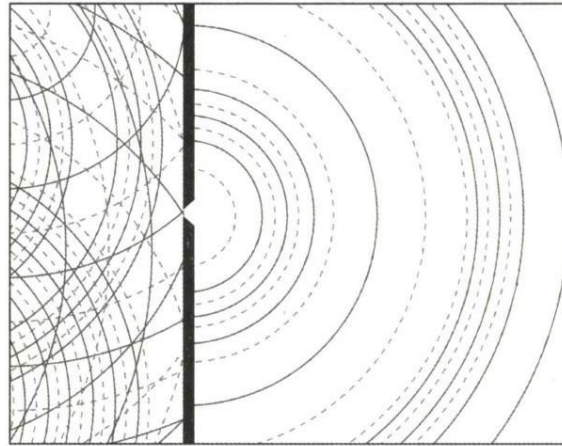


Fresnel-Arago laws:

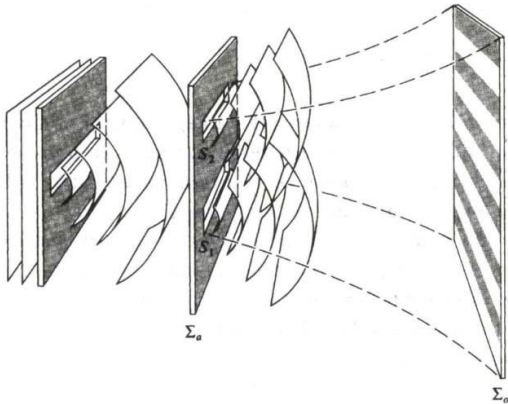
- ❑ Two orthogonal, coherent $\wp$ -states cannot interfere in the sense that $I_{12} = 0$ and no fringes result.
- ❑ Two parallel, coherent $\wp$ -state will interfere in the same way as will natural light.
- ❑ The two constituent orthogonal $\wp$ -states of natural light cannot interfere to form a readily observable fringe pattern even if rotated into alignment. Since these $\wp$ -states are incoherent.

8.3 Wavefront-splitting Interferometers

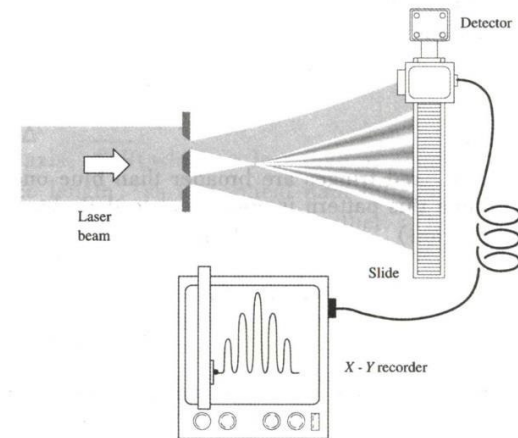
Young's experiment



Classic arrangement of Young's experiment

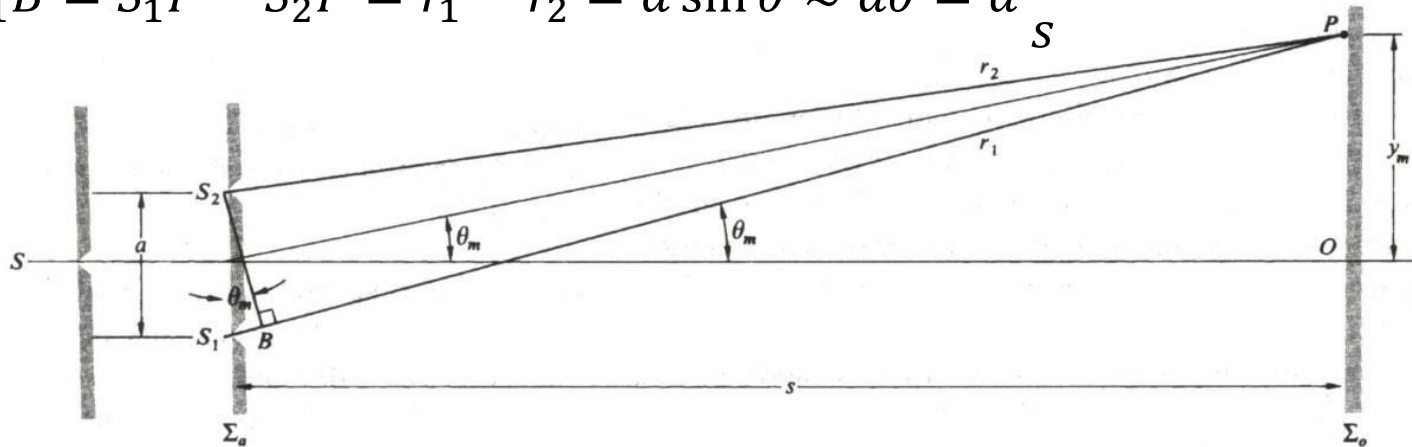


A modern version



The optical path difference

$$\overline{S_1B} = \overline{S_1P} - \overline{S_2P} = r_1 - r_2 = a \sin \theta \approx a\theta = a \frac{y}{s}$$



Constructive interference will occur when $r_1 - r_2 = m\lambda$

The position of the m-th bright fringe on the screen (the maximum at 0 is the zeroth fringe)

$$y_m \approx \frac{s}{a} m\lambda$$

The angular position of the m-th bright fringe: $\theta_m \approx \frac{y_m}{s} = \frac{m\lambda}{a}$

The spacing of the fringes on the screen:

$$\Delta y = y_{m+1} - y_m \approx \frac{s}{a} (m+1)\lambda - \frac{s}{a} m\lambda = \frac{s\lambda}{a}$$

Note: Red fringes are broader than blue ones.

Provided the two beams: coherence; equal irradiances

$$I = 4I_0 \cos^2 \frac{k(r_1 - r_2)}{2}$$

Phase difference:

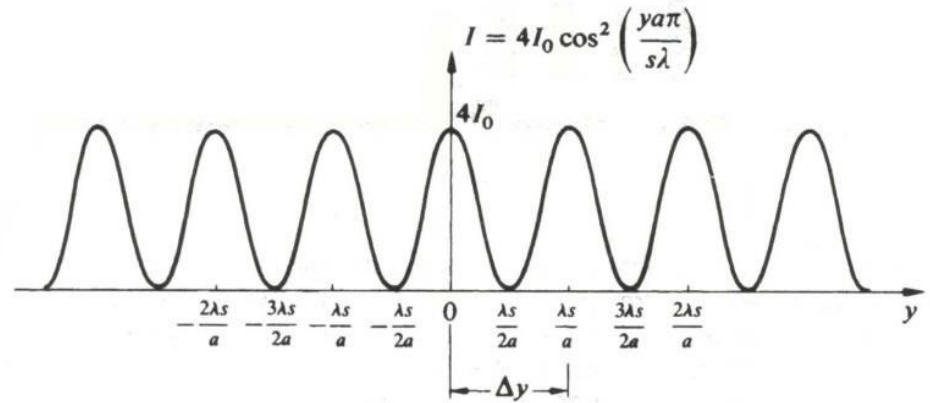
$$\delta = k(r_1 - r_2)$$

Optical path difference:

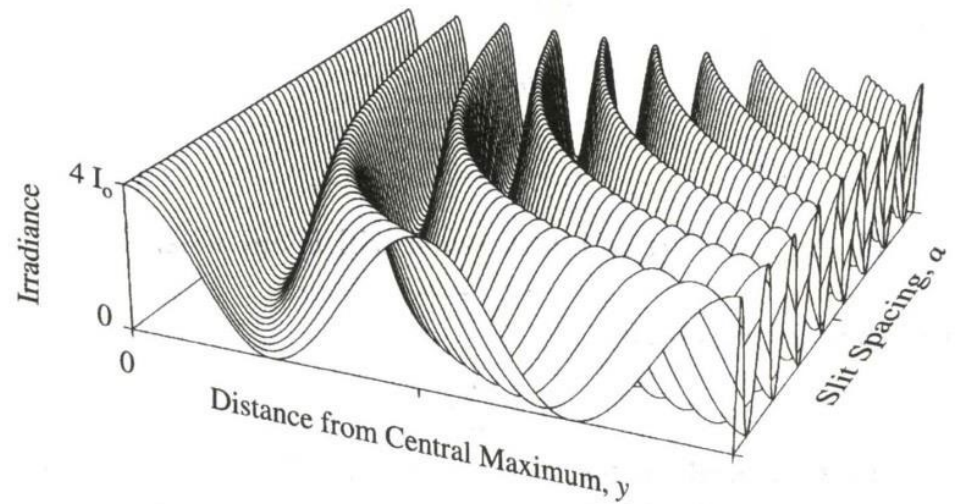
$$r_1 - r_2 = a \frac{y}{s}$$

The resultant irradiance:

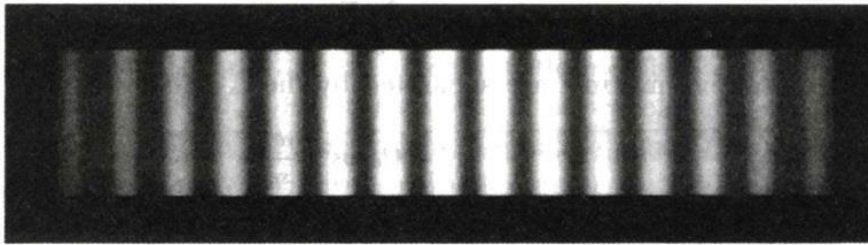
$$I = 4I_0 \cos^2 \frac{ya\pi}{s\lambda}$$



(a)



(b)



Young's double slit experiment

The effect of finite coherent length

Consider the point further above or below the axis, the S_1B will increase.

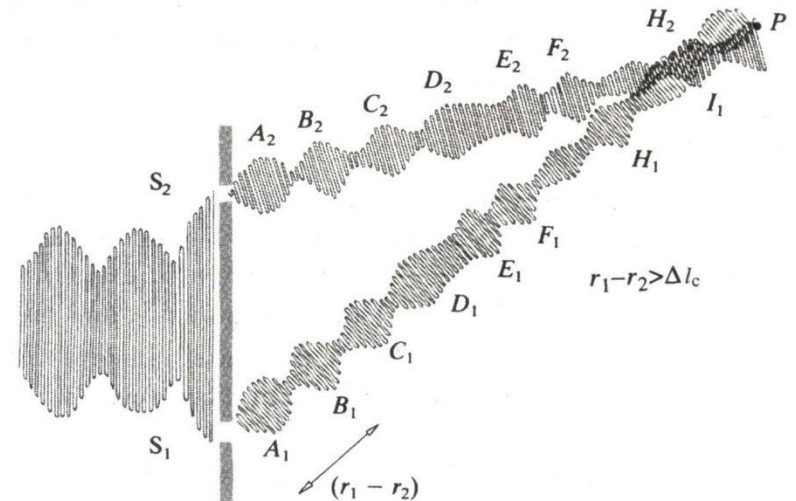
Short coherence length:

- ❑ Identically paired wavegroups will not arrive at P exactly together: the contrast of the fringes will degrade.

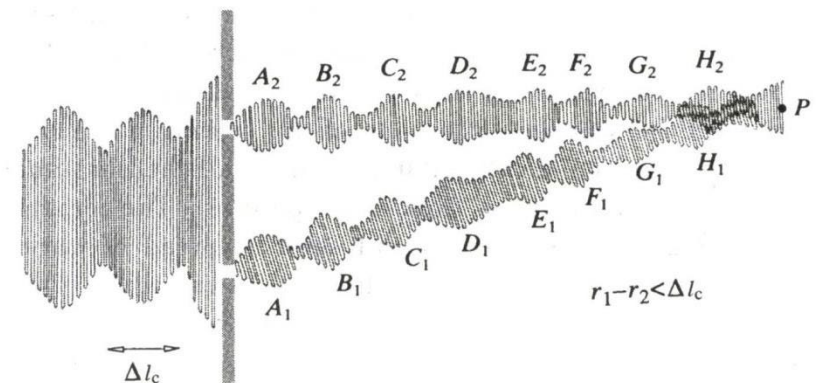
White light:

- ❑ All the constituent colors will arrive at $y=0$ in phase, so the zeroth-order fringe will be essentially white.
- ❑ All other higher order maxima will show a spread of wavelengths, since

$$y_m = \frac{s}{a} m \lambda.$$



(a)



(b)